Data Management

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Setup

```
# Load packages
pacman::p_load(tidyverse, pacman, grateful, here, usethis)
```

```
# Set blank slate
usethis::use_blank_slate(scope = "user")

# Citations
grateful::cite_packages(pkgs='Session', output='paragraph', out.dir='.')
```

We used R v. 4.5.2 [@base] and the following R packages: here v. 1.0.2 [@here], pacman v. 0.5.1 [@pacman], tidyverse v. 2.0.0 [@tidyverse], usethis v. 3.2.2 [@usethis].

```
print(paste("operating system: ", sessionInfo()$running))
```

```
[1] "operating system:  macOS Tahoe 26.5.2"
```

```
# Check working directory
here()
```

```
[1] "/Users/kieramurray/Library/CloudStorage/OneDrive-Personal/Grad School/Biostats in Actio
```

Data processing

Load raw data

```r
# Load in the raw data
load(here("data", "raw", "acupuncture_data_reduced.RData"))
acupunc_full <- data
```

Recode categorical variables as factor

```r
# sex: 2 values, 1=female, 0=male
acupunc_full$sex <- factor(data$sex,
                           levels = c(1, 0),
                           labels = c("female", "male"))

# migraine: 2 values, 1=migraine, 0=tension-type
acupunc_full$migraine <- factor(data$migraine,
                                levels = c(1, 0),
                                labels = c("migraine", "tension-type"))

# group: 2 values, 1=acupuncture, 0=control
acupunc_full$group <- factor(data$group,
                             levels = c(1, 0),
                             labels = c("acupuncture", "control"))

# acupuncturist: 12 unique values, numbered 1 to 12
acupuncturist_int <- seq(min(data$acupuncturist),
                         max(data$acupuncturist)) # vector of numeric values
acupuncturist_str <- as.character(acupuncturist_int) # vector of character values
acupunc_full$acupuncturist <- factor(data$acupuncturist,
                                     levels = acupuncturist_int,
                                     labels = acupuncturist_str)

# GP: 36 unique integer values; not numbered in an obvious way
gp_int <- seq(min(data$practice_id),
              max(data$practice_id)) # vector of potential numeric values
gp_str <- as.character(gp_int) # vector of potential character values
acupunc_full$gp <- factor(data$practice_id,
                          levels = gp_int,
                          labels = gp_str)

# patient ID: 401 unique integer values; not numbered in an obvious way
id_int <- seq(min(data$id),
              max(data$id)) # vector of potential numeric values
```

```
id_str <- as.character(id_int) # vector of potential character values
acupunc_full$id <- factor(data$id,
                          levels = id_int,
                          labels = id_str)
```

Save the full recoded data

```
# Drop redundant variable practice_id (the numeric version of GP)
acupunc_full <- acupunc_full %>% select(-practice_id)

# Check the recoded data
summary(acupunc_full)
```

```
      id           age            sex              migraine     chronicity
 100    :  1   Min.   :18.00   female:337   migraine    :377   Min.   : 1.00
 101    :  1   1st Qu.:39.00   male  : 64   tension-type: 24   1st Qu.:10.00
 104    :  1   Median :48.00                                   Median :20.00
 105    :  1   Mean   :45.54                                   Mean   :21.46
 108    :  1   3rd Qu.:53.00                                   3rd Qu.:31.00
 112    :  1   Max.   :65.00                                   Max.   :54.00
 (Other):395
 acupuncturist         group          pk1             pk2
 3      :91    acupuncture:205   Min.   : 6.75   Min.   : 0.00
 8      :61    control    :196   1st Qu.:15.00   1st Qu.:10.75
 6      :56                      Median :21.00   Median :16.88
 7      :38                      Mean   :26.51   Mean   :21.60
 9      :36                      3rd Qu.:34.25   3rd Qu.:27.00
 11     :34                      Max.   :94.75   Max.   :98.50
 (Other):85                                      NA's   :75
     pk5              f1              f2              f5               gp
 Min.   : 0.00   Min.   : 3.00   Min.   : 0.00   Min.   : 0.000   18     : 43
 1st Qu.: 8.50   1st Qu.:10.00   1st Qu.: 4.00   1st Qu.: 0.000   12     : 37
 Median :15.00   Median :15.00   Median :10.00   Median : 8.000   41     : 32
 Mean   :19.08   Mean   :16.18   Mean   :10.94   Mean   : 9.319   10     : 24
 3rd Qu.:25.25   3rd Qu.:21.00   3rd Qu.:17.00   3rd Qu.:14.000   16     : 22
 Max.   :87.25   Max.   :28.00   Max.   :28.00   Max.   :28.000   40     : 22
 NA's   :100                                                      (Other):221
```

```
# Save as .RData to data/processed
save(acupunc_full, file = here("data", "processed", "acupunc_full.RData"))
```

Save the migraine-only recoded data

```
# Filter out tension-type patients
acupunc <- acupunc_full %>% filter(migraine == "migraine")

# Check the filtered data - this will be used for analysis
summary(acupunc)
```

```
       id           age            sex              migraine     chronicity
 100    :  1   Min.   :18.00   female:318   migraine    :377   Min.   : 1.00
 101    :  1   1st Qu.:38.00   male  : 59   tension-type:  0   1st Qu.:10.00
 104    :  1   Median :48.00                                   Median :20.00
 105    :  1   Mean   :45.42                                   Mean   :21.86
 108    :  1   3rd Qu.:53.00                                   3rd Qu.:31.00
 112    :  1   Max.   :65.00                                   Max.   :54.00
 (Other):371
 acupuncturist         group          pk1              pk2
 3      :83    acupuncture:194   Min.   : 6.75   Min.   : 0.00
 8      :58    control    :183   1st Qu.:15.00   1st Qu.:10.75
 6      :54                      Median :21.25   Median :16.75
 7      :37                      Mean   :26.61   Mean   :21.42
 9      :34                      3rd Qu.:34.50   3rd Qu.:27.06
 11     :32                      Max.   :94.75   Max.   :98.50
 (Other):79                                      NA's   :69
      pk5              f1              f2              f5               gp
 Min.   : 0.00   Min.   : 3.00   Min.   : 0.00   Min.   : 0.000   18     : 41
 1st Qu.: 8.25   1st Qu.:10.00   1st Qu.: 4.00   1st Qu.: 0.000   12     : 31
 Median :15.00   Median :15.00   Median :10.00   Median : 8.000   41     : 30
 Mean   :18.96   Mean   :16.14   Mean   :10.96   Mean   : 9.265   10     : 23
 3rd Qu.:25.06   3rd Qu.:21.00   3rd Qu.:17.00   3rd Qu.:14.000   16     : 22
 Max.   :87.25   Max.   :28.00   Max.   :28.00   Max.   :28.000   40     : 20
 NA's   :93                                                       (Other):210
```

```
# Save as .RData to data/processed
save(acupunc, file = here("data", "processed", "acupunc.RData"))
```